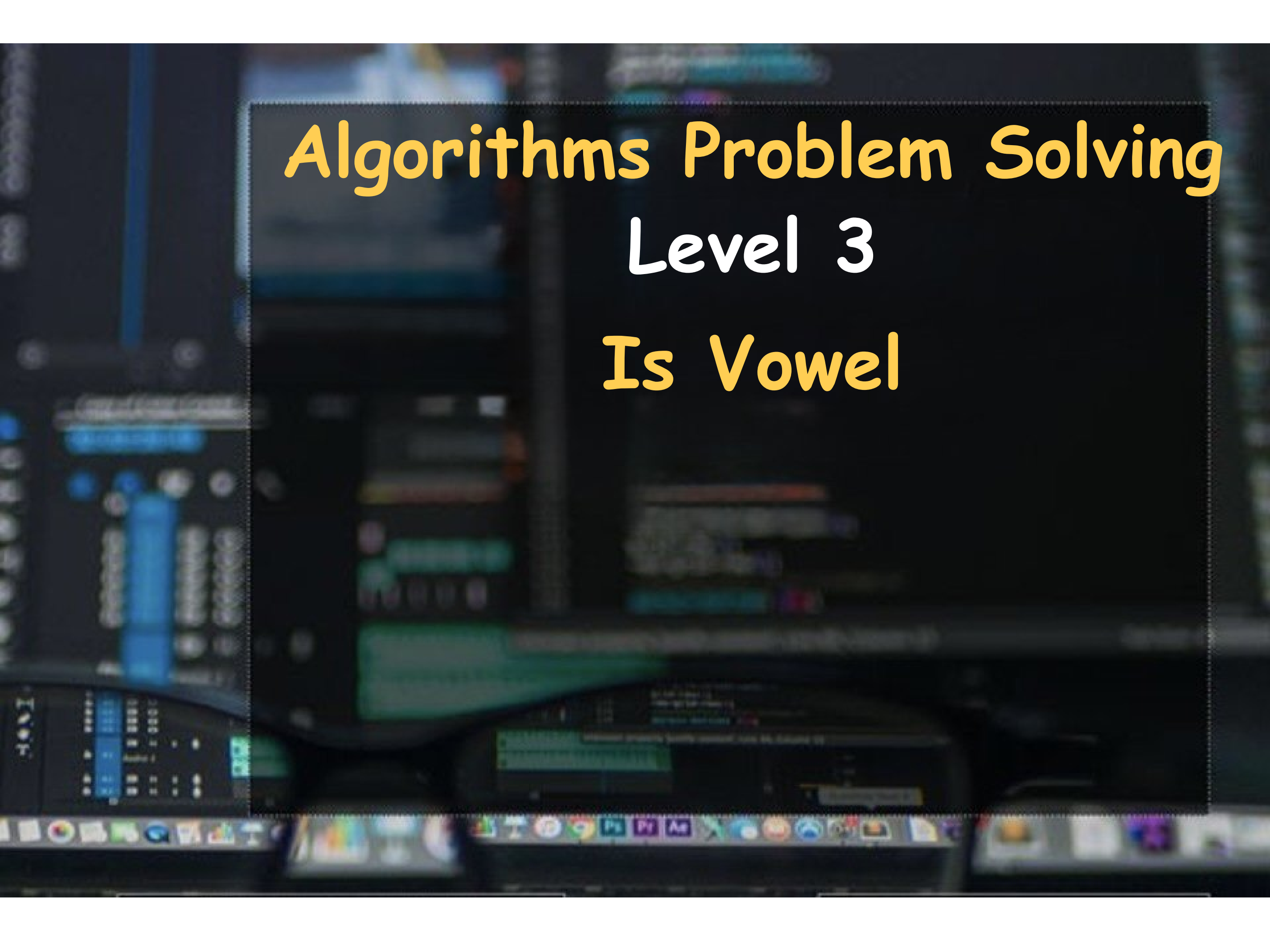




Algorithms Problem Solving Level 3 Is Vowel



Algorithms Problem Solving

Level 3

Is Vowel

Definition

Atomic Vowel Validation

-  Program determines whether one character is a vowel.

-  The function returns ``true`` for ``a``, ``e``, ``i``, ``o``, or ``u``, regardless of case.

Explanation

Case Normalization

-  The input character can be converted to lowercase using ``tolower()``.
-  This avoids separately checking uppercase vowels.

Vowel Membership Check

-  The normalized character is compared against the five vowel values.
-  The comparisons are combined using logical OR.

Boolean Result

-  The function returns ``true`` when the character belongs to the vowel set.
-  Otherwise it returns ``false``.

Procedures

- 1 Read one character.
- 2 Pass it to ``IsVowel()'`.
- 3 Normalize its case.
- 4 Check the vowel conditions.
- 5 Print the boolean evaluation.

💡 Example

◆ Input: a

➡ Result

YES: Letter 'a' is Vowel

◆ Input: b

➡ Result

NO: Letter 'b' is NOT Vowel

Note

- ◆ A helper predicate can be reused by `CountVowels()` and `PrintVowels()`.
- ◆ Returning `bool` makes the function convenient inside `if` conditions.
- ◆ The function checks membership in a fixed character set.

⊘ Warning

- ◆ ``cin >> ch`` reads only one non-whitespace character.
- ◆ Without case normalization, uppercase vowels must be checked separately.
- ◆ The function should not be confused with checking whether a complete string contains a vowel.



Comparison

▨ `IsVowel(char)`

- ✓ Encapsulates vowel logic in one reusable function.
- ✓ Returns a boolean.
- ✓ Can be called from multiple string-processing functions.

▤ Inline Vowel Condition

- ✓ Places all vowel comparisons directly inside another function.
- ✓ Avoids a helper function.
- ✓ Can become repetitive when multiple algorithms need the same test.

✦✦ Summary

- ✓ Receives one character.
- ✓ Normalizes its case.
- ✓ Checks membership in the vowel set.
- ✓ Returns a reusable boolean result.

Thank you 😊